

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the integral $\int_0^2 e^{-x^3} dx$.

a) Find a series solution to the definite integral $\int_0^2 e^{-x^3} dx$.

$$\int_0^2 e^{-x^3} dx = \int_0^2 \sum_{n=0}^{\infty} \frac{(-x^3)^n}{n!} dx = \int_0^2 \sum_{n=0}^{\infty} \frac{(-1)^n x^{3n}}{n!} dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{3n+1}}{(3n+1)n!} \bigg|_0^2 = \sum_{n=0}^{\infty} \frac{(-1)^n (2)^{3n+1}}{(3n+1)n!}.$$

b) How did you know that you could evaluate the series at $x=2$ & $x=0$?

The IOC for e^{-x^2} is $(-\infty, \infty)$ and since $x=2$ & $x=0$ are within the IOC it is okay to evaluate the series at these values and maintain equality.

2. Consider the value S given by $S = \frac{1}{e^0 \cdot 0!} + \frac{2}{e^1 \cdot 1!} + \frac{4}{e^2 \cdot 2!} + \dots$. Determine $\ln(S)$.

Note that the given series can be expressed as $S = \sum_{n=0}^{\infty} \frac{2^n}{e^n \cdot n!} = \sum_{n=0}^{\infty} \frac{\left(\frac{2}{e}\right)^n}{n!} = e^{2/e}$.

Therefore, $\ln(S) = \ln\left(e^{2/e}\right) = \frac{2}{e}$.

3. Consider the function $f(x) = 2x \ln(1+x^2)$

a) Determine a power series for $f(x)$ about $x=0$.

We know that $\ln(x+1) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}$ on $(-1,1]$.

If we substitute x^2 in place of x we obtain $\ln(x^2+1) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{n+1}}{n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{n+1}$. Since the original series was convergent for $-1 < x \leq 1$ we have $0 \leq x^2 \leq 1 \rightarrow 0 \leq |x| \leq 1 \rightarrow -1 \leq x \leq 1$.

If we then multiply by $2x$ we obtain $2x \ln(x^2+1) = \sum_{n=0}^{\infty} \frac{2(-1)^n x^{2n+3}}{n+1}$ on the interval $[-1,1]$.

b) Based on the power series, what should the 13th derivative of $f(x)$ be at $x=0$?

We know that the coefficients of a Taylor series centered at $x=0$ are defined as $c_n = \frac{f^{(n)}(0)}{n!}$.

From part (a) we have coefficients in the form $c_n = \frac{2(-1)^n}{n+1}$ and if $n=5$ then we have the term with x^{13} .

Therefore $\frac{f^{(13)}(0)}{13!} = \frac{2(-1)^5}{5+1} \rightarrow f^{(13)}(0) = \frac{-2 \cdot 13!}{6} = -\frac{13!}{3}$

c) Determine the exact value of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n+2}(n+1)}$.

This series looks like the series we found, just being evaluated at $x = \frac{1}{2}$. Since $x = \frac{1}{2}$ is within the IOC

of $[-1,1]$ we know that $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n+2}(n+1)} = \sum_{n=0}^{\infty} \frac{2(-1)^n \left(\frac{1}{2}\right)^{2n+3}}{n+1} = 2\left(\frac{1}{2}\right) \ln\left(\left(\frac{1}{2}\right)^2 + 1\right) = \ln(1.25)$

4. Determine the exact value of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n \cdot 2^n}$.

We know that $\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$ with an IOC $(-1, 1]$.

If we evaluate at $x = \frac{1}{2}$ we see that $\ln(1.5) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \left(\frac{1}{2}\right)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n \cdot 2^n}$.

5. Consider the function $f(x) = \frac{1}{(1+x)^2}$.

- a) Use differentiation to help find a power series representation for this function. What is the radius of convergence for this power series?

We know that $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n$ with $R=1$.

Then, by differentiating we have that

$$\begin{aligned} f(x) &= \frac{d}{dx} \left[\frac{1}{1+x} \right] = \frac{d}{dx} \left[\sum_{n=0}^{\infty} (-1)^n x^n \right] \\ &\rightarrow \frac{1}{(1+x)^2} = \sum_{n=1}^{\infty} (-1)^n n x^{n-1} \end{aligned}$$

- b) Use your answer from part (a) to find a power series representation for $g(x) = \frac{1}{(1+x)^3}$ and state its radius of convergence.

By differentiating again we have that

$$\begin{aligned} \frac{d}{dx} \left[\frac{1}{(1+x)^2} \right] &= \frac{d}{dx} \left[\sum_{n=1}^{\infty} (-1)^n n x^{n-1} \right] \\ &\rightarrow \frac{-2}{(1+x)^3} = \sum_{n=1}^{\infty} (-1)^n n(n-1) x^{n-2} \quad w/ \quad R=1 \end{aligned}$$

Now we just divide both sides by -2 and get

$$\frac{1}{(1+x)^3} = -\frac{1}{2} \sum_{n=1}^{\infty} (-1)^n n(n-1) x^{n-2} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{2} (n-1) x^{n-2} \quad w/ \quad R=1$$

- c) Use your answer from part (b) to find a power series representation for $h(x) = \frac{x^2}{(1+x)^3}$ and state its radius of convergence.

We know from part (b) that $\frac{1}{(1+x)^3} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{2} (n-1) x^{n-2}$ w/ $R=1$.

Now we just multiply both sides by x^2 .

$$\frac{x^2}{(1+x)^3} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{2} (n-1) x \quad \text{w/ } R=1$$

- d) Use your answer from part (a) to find a power series representation for $k(x) = \frac{1}{(1+2x)^2}$ and state its radius of convergence.

From part (a) we know that $\frac{1}{(1+x)^2} = \sum_{n=1}^{\infty} (-1)^n n x^{n-1}$ with $R=1$.

We will replace x with $2x$. $\frac{1}{(1+2x)^2} = \sum_{n=1}^{\infty} (-1)^n n (2x)^{n-1}$.

We know that the original IOC was, at least, $-1 < x < 1$. Since we replaced x with $2x$ we now have $-1 < 2x < 1 \rightarrow -\frac{1}{2} < x < \frac{1}{2}$. Therefore, the new ROC would be $R = \frac{1}{2}$.

6. Find a function that is modeled by each of the following power series. Determine the radius of convergence for each of these power series.

a) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{6n+4}}{(2n+1)!}$

We note that the basic shape of this series resembles the series for the sine function,

$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$. By rewriting the series we see that

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{6n+4}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (x^3)^{2n+1} \cdot x}{(2n+1)!} = -x \sin(x^3).$$

The radius of convergence for the series for $\sin(x)$ is $R = \infty$. This implies that $-\infty < x < \infty$. We note that with the substitution of x^3 in place of x this makes $-\infty < x^3 < \infty \rightarrow -\infty < x < \infty$. Therefore, the radius of convergence is still $R = \infty$.

$$b) \sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+4}}{(2n+1)}$$

We note that the basic shape of this series resembles the series for the inverse tangent function,

$$\tan^{-1}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)}. \text{ By rewriting the series we see that}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+4}}{(2n+1)} = \sum_{n=0}^{\infty} \frac{(-1)^n (3x^2)^{2n+1} \cdot x^2}{(2n+1)} = x^2 \tan^{-1}(3x^2).$$

The radius of convergence for the series for $\tan^{-1}(x)$ is $R=1$ with IOC $[-1,1]$. This implies that

$-1 \leq x \leq 1$. We note that with the substitution of $3x^2$ in place of x this makes

$$-1 \leq 3x^2 \leq 1 \rightarrow 0 \leq 3x^2 \leq 1 \rightarrow 0 \leq x^2 \leq \frac{1}{3} \rightarrow -\frac{1}{\sqrt{3}} \leq x \leq \frac{1}{\sqrt{3}}. \text{ Therefore, the radius of convergence}$$

$$\text{is } R = \frac{1}{\sqrt{3}}.$$

$$c) \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+2}}{(n+1)(n+2)}$$

We note that the basic shape of this series resembles the series for the natural logarithm,

$$\ln(x+1) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}. \text{ If we replace } x \text{ with } 5x \text{ we see that}$$

$$\ln(x+1) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} \rightarrow \ln(5x+1) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{n+1}}{n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+1}}{n+1}.$$

Now we just need to take the antiderivative of this series and we will have the given series.

This means that

$$\int \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+1}}{n+1} dx = \int \ln(5x+1) dx$$

$$C + \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+2}}{(n+1)(n+2)} dx = \int \ln(5x+1) dx$$

To integrate the logarithm we use IBP.

$$\text{Let } u = \ln(5x+1) \text{ \& } dv = dx, \text{ then } u = \frac{5}{5x+1} \text{ \& } v = x. \text{ Therefore,}$$

$$\int \ln(5x+1) dx = \ln(5x+1)x - \int \frac{5x}{5x+1} dx.$$

On the new integrand we perform a polynomial long division or use some clever algebra.

Polynomial long division

$$\begin{array}{r} 1 \\ (5x+1) \overline{) 5x+0} \\ \underline{-(5x+1)} \\ -1 \end{array}$$

$$\text{Therefore, } \int \frac{5x}{5x+1} dx = \int \left(1 - \frac{1}{5x+1} \right) dx$$

Clever algebra

$$\frac{5x}{5x+1} = \frac{5x+1-1}{5x+1} = \frac{5x+1}{5x+1} - \frac{1}{5x+1} = 1 - \frac{1}{5x+1}$$

So, now we have that

$$\begin{aligned} \int \ln(5x+1) dx &= \ln(5x+1)x - \int \frac{5x}{5x+1} dx \\ &= \ln(5x+1)x - \int \left(1 - \frac{1}{5x+1} \right) dx \\ &= \ln(5x+1)x - \left(x - \frac{1}{5} \ln|5x+1| \right) + C \\ &= \ln(5x+1)x - x + \frac{1}{5} \ln|5x+1| + C \end{aligned}$$

Thus we have established that

$$C + \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+2}}{(n+1)(n+2)} dx = \int \ln(5x+1) dx \rightarrow C + \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+2}}{(n+1)(n+2)} dx = \ln(5x+1)x - x + \frac{1}{5} \ln|5x+1|$$

If we let $x=0$ (this value is in the resulting IOC) we see that

$$C + \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} (0)^{n+2}}{(n+1)(n+2)} dx = \ln(5(0)+1)(0) - (0) + \frac{1}{5} \ln|5(0)+1| \rightarrow C = 0.$$

$$\text{Therefore, } \sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+2}}{(n+1)(n+2)} dx = \ln(5x+1)x - x + \frac{1}{5} \ln|5x+1|.$$

The radius of convergence for the series for $\ln(x+1)$ is $R=1$ with IOC $(-1,1)$. This implies that

$$-1 < x < 1. \text{ We note that with the substitution of } 5x \text{ in place of } x \text{ this makes } -1 < 5x < 1 \rightarrow -\frac{1}{5} < x < \frac{1}{5}.$$

Therefore, the radius of convergence is $R = \frac{1}{5}$. After taking the antiderivative the ROC doesn't change

so we still have $R = \frac{1}{5}$.

7. Consider the function $f(x) = \int_0^x \frac{\tan^{-1}(t)}{t} dt$. It can be shown that the discontinuity at $x=0$ is not a problem and the improper integral will converge.

a) Determine the Taylor series for the function $f(x) = \int_0^x \frac{\tan^{-1}(t)}{t} dt$.

Note that $\tan^{-1}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$ for x -values in $[-1,1]$.

Therefore, $\frac{\tan^{-1}(t)}{t} = \frac{1}{t} \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{2n+1}$.

So, $f(x) = \int_0^x \frac{\tan^{-1}(t)}{t} dt = \int_0^x \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{2n+1} dt = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)^2} \Big|_0^x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)^2}$ for x -values in $[-1,1]$.

b) Use your answer from part (a) to determine the value of $\int_0^1 \frac{\tan^{-1}(t)}{t} dt$ accurately to 2 decimal places.

From part (a) we know that since $x=1$ is inside of the IOC for the power series that

$$\int_0^1 \frac{\tan^{-1}(t)}{t} dt = f(1) = \sum_{n=0}^{\infty} \frac{(-1)^n (1)^{2n+1}}{(2n+1)^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}.$$

This is an alternating series and therefore to estimate the exact answer to 2 decimal places we need to take a partial sum of the series using n terms where $|S - s_n| \leq a_{n+1} \leq \frac{1}{100}$.

Note that $a_n = \frac{1}{(2n+1)^2}$. So we must find n such that $\frac{1}{(2(n+1)+1)^2} \leq \frac{1}{100}$.

We calculate $\frac{1}{(2(n+1)+1)^2} \leq \frac{1}{100} \rightarrow 100 \leq (2n+3)^2 \rightarrow 10 \leq 2n+3 \rightarrow n \geq \frac{7}{2}$.

Therefore $n \geq 4$.

So, we have that $\int_0^1 \frac{\tan^{-1}(t)}{t} dt \approx \frac{1}{1} - \frac{1}{9} + \frac{1}{25} - \frac{1}{81} \approx 0.91$

8. Power series can be used to solve differential equations. Loosely speaking a differential equation is an equation that relate derivatives of a given function. Suppose that a country has at time t a population of $p(t)$, and that the rate of change of the population is given by $p'(t) = 2p(t)$. Here, the equation $p'(t) = 2p(t)$ is a differential equation. Solving this equation entails determining what function(s) $p(t)$ could satisfy this equation.

a) If we represent the total population of the country by a power series $p(t) = p(0) + p(1)t + p(2)t^2 + \dots$, then use this power series and the differential equation to find a relationship between the coefficients of the power series. (Hint: Think about how you can re-express the differential equation using the power series)

$$\text{We have } p(t) = p(0) + p(1)t + p(2)t^2 + p(3)t^3 + p(4)t^4 + \dots$$

$$\text{So, } p'(t) = p(1) + 2p(2)t + 3p(3)t^2 + 4p(4)t^3 + \dots$$

So, the differential equation can be written as

$$p'(t) = 2p(t)$$

$$p(1) + 2p(2)t + 3p(3)t^2 + 4p(4)t^3 + \dots = 2[p(0) + p(1)t + p(2)t^2 + p(3)t^3 + p(4)t^4 + \dots]$$

$$p(1) + 2p(2)t + 3p(3)t^2 + 4p(4)t^3 + \dots = 2p(0) + 2p(1)t + 2p(2)t^2 + 2p(3)t^3 + 2p(4)t^4 + \dots$$

In order for these two sides to be equal it must be the case that

$$p(1) = 2p(0)$$

$$2p(2) = 2p(1)$$

$$3p(3) = 2p(2)$$

$$4p(4) = 2p(3)$$

From here we can establish the pattern of $n \cdot p(n) = 2p(n-1) \rightarrow p(n) = \frac{2}{n} p(n-1)$.

b) Find the power series if $p(0) = 1$. (Hint: Use your knowledge from part (a) to find the coefficientwise of the power series. These coefficients can be expressed in a way that involves powers of 2 and factorials.)

We only need to find the coefficients of the power series $p(0), p(1), p(2), \dots$

We established in part (a) that $p(n) = \frac{2}{n} p(n-1)$. If $p(0) = 1$ then,

$$p(1) = \frac{2}{1} p(0) = 2(1) = 2 = \frac{2}{1} = \frac{2}{1!}$$

$$p(2) = \frac{2}{2} p(1) = 1(2) = 2 = \frac{4}{2} = \frac{2^2}{2!}$$

$$p(3) = \frac{2}{3} p(2) = \frac{2}{3}(2) = \frac{4}{3} = \frac{8}{6} = \frac{2^3}{3!}$$

$$p(4) = \frac{2}{4} p(3) = \frac{1}{2} \left(\frac{4}{3} \right) = \frac{2}{3} = \frac{16}{24} = \frac{2^4}{4!}$$

Then the power series can be expressed as $\sum_{n=0}^{\infty} \frac{2^n}{n!} t^n$

c) Identify the power series as the Taylor expansion of some function $f(t)$.

$$\sum_{n=0}^{\infty} \frac{2^n}{n!} t^n = \sum_{n=0}^{\infty} \frac{(2t)^n}{n!} = e^{2t}$$

9. Use Maclaurin series (not l'Hopital's rule) to find the following important limits.

a) $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$

We know that $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$ and so

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(x)}{x} &= \lim_{x \rightarrow 0} \frac{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}}{x} = \lim_{x \rightarrow 0} \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = \lim_{x \rightarrow 0} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n} \\ &= \lim_{x \rightarrow 0} \left[1 - \frac{1}{3!} x^2 + \frac{1}{5!} x^4 - \frac{1}{7!} x^6 + \dots \right] \\ &= 1 \end{aligned}$$

b) $\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x}$

We know that $\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x} &= \lim_{x \rightarrow 0} \frac{1 - \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}}{x} = \lim_{x \rightarrow 0} \frac{1 - \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right]}{x} = \lim_{x \rightarrow 0} \frac{\left[\frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} - \dots \right]}{x} \\ &= \lim_{x \rightarrow 0} \left[\frac{x}{2!} - \frac{x^3}{4!} + \frac{x^5}{6!} - \dots \right] \\ &= 0 \end{aligned}$$